

DDAE: Dual-Flow Adaptive Diffusion Autoencoder for High-Fidelity Image Reconstruction and Editing

Abstract

Diffusion autoencoders have shown strong potential for image reconstruction, representation learning, and semantic editing, yet they still suffer from unstable reconstruction and degraded generation quality when score estimation becomes unreliable at different noise levels. This paper proposes **Dual-Flow Adaptive Diffusion Autoencoder (DDAE)**, a diffusion-based autoencoding framework that improves reconstruction and editing stability through decoupled representations, adaptive Dual-Flow score estimation, and implicit constraints on the stochastic encoder. Instead of relying on a single prediction path, DDAE jointly exploits image-space and noise-space estimations and adaptively fuses them at different denoising timesteps, allowing the model to benefit from their complementary advantages under varying signal-to-noise ratios. To further reduce inconsistency between the stochastic encoding trajectory and the generation trajectory, we introduce an implicit constraint mechanism that aligns intermediate representations through attention-based feature guidance. Experiments on CelebA-HQ and iLab-20M demonstrate that DDAE achieves competitive or superior reconstruction quality compared with existing diffusion autoencoder and generative baselines. The proposed method also produces smooth interpolation trajectories and effective image editing results while preserving semantic consistency. These results indicate that adaptive Dual-Flow score matching and implicit stochastic-encoder constraints provide an effective strategy for stable, high-fidelity diffusion autoencoding.

Keywords: Diffusion models; Autoencoders; Image reconstruction and editing; Score estimation

1. Introduction

Diffusion probabilistic models learn image generation by reversing a gradual Gaussian corruption process [Ho et al. \(2020\)](#); [Nichol and Dhariwal \(2021\)](#); [Dhariwal and Nichol \(2021\)](#); [Rombach et al. \(2022\)](#). Deterministic samplers such as DDIM make the sampling trajectory usable for interpolation, inversion, and image editing [Song et al. \(2021a\)](#), while score-based SDE formulations provide a continuous-time interpretation of denoising and probability-flow ODE sampling [Song et al. \(2021b\)](#). These developments have made diffusion models strong decoders for image reconstruction and editing. However, standard diffusion models do not expose a compact semantic code. Diffusion autoencoders address this limitation by pairing a semantic encoder with a conditional diffusion decoder [Preechakul et al. \(2022\)](#); [Zhang et al. \(2022\)](#), but their reconstruction quality still depends heavily on how the reverse transition is estimated and how accurately the terminal noise state is inverted.

We study these two sources of error. Firstly, reverse-process estimation is often implemented through a single prediction target, most commonly the noise ϵ or the clean image x_0 . These targets are algebraically related, but they behave differently across the signal-to-noise schedule. When the signal is weak, a small noise-prediction error can be strongly amplified when converted into a clean-image estimate. Conversely, direct x_0 prediction is close to a

reconstruction objective and can provide weaker denoising supervision in lightly corrupted states. This suggests that the two targets are complementary rather than interchangeable.

Secondly, deterministic inversion is only approximate. In a diffusion autoencoder, DDIM inversion is used as a stochastic encoder that maps an input image to a terminal state x_T^* . The subsequent generation process starts from this state and is expected to return to the input image under the same semantic code. In practice, inversion and generation follow two trajectories that share the learned vector field only approximately, and their mismatch accumulates over denoising steps. This is particularly harmful for high-frequency details such as vehicle boundaries, facial texture, and background patterns.

We propose **Dual-Flow Adaptive Diffusion Autoencoder (DDAE)** to address these limitations. DDAE estimates the reverse posterior mean through both an x_0 -prediction flow and an ϵ -prediction flow. The two mean estimates are fused in posterior-mean space by a learned gate, allowing the decoder to use the more reliable estimate at each timestep. We also introduce an implicit consistency constraint for deterministic reconstruction: during DDIM inversion, DDAE stores late-stage self-attention key-value features, and during generation it reuses these features in selected decoder layers. This aligns local appearance details without adding trainable parameters or changing the sampler.

DDAE is designed for both reconstruction and editing. For reconstruction, the semantic encoder and inferred terminal noise provide a two-part code. For editing, the semantic code can be processed by a group-supervised autoencoder (GAE) to obtain disentangled groups corresponding to identity, pose, and background [Leng et al. \(2023\)](#); [Ge et al. \(2021\)](#). The edited semantic code is then decoded by DDAE, preserving image quality while enabling attribute-level control.

We make three main contributions:

- We analyze two practical failure modes of diffusion autoencoders: target-dependent reverse estimation error and inversion-generation trajectory mismatch.
- We propose Dual-Flow adaptive posterior estimation, which fuses x_0 - and ϵ -based posterior means with a learned gate that conditions on both timestep and semantic latent. Unlike discrete prediction-target switching, this enables smooth per-timestep interpolation in posterior-mean space.
- We introduce a parameter-free implicit consistency constraint based on self-attention key-value reuse, and validate DDAE on iLab-20M, FFHQ, and CelebA-HQ with quantitative reconstruction metrics and qualitative editing results.

2. Related Work

2.1. Diffusion Models and Latent-Space Autoencoding

Diffusion models have emerged as one of the most prominent architectures in visual generation, with the core idea of learning high-quality image synthesis through iterative noise injection and denoising. Latent Diffusion Models (LDM) perform the diffusion process in a *latent space*, achieving more efficient generation and being widely adopted in systems such as Stable Diffusion. Under latent-space guidance, diffusion autoencoders have been proposed to obtain interpretable and manipulable low-dimensional latent variables, thereby enhancing

editability. For example, DiffuseGAE improves latent-space disentanglement via group-supervised learning, enabling multi-attribute high-fidelity image manipulation [Lu et al. \(2023\)](#). Furthermore, Hierarchical Diffusion Autoencoders (HDAE) achieve finer-grained image editing and blending through hierarchical feature truncation [Lu et al. \(2024\)](#). These studies provide an important methodological foundation for DDAE’s latent-space design.

2.2. Representation Learning and Diffusion Models

The synergy between diffusion models and representation learning has become increasingly apparent. On one hand, diffusion models possess inherent self-supervised properties, making them effective for unsupervised representation learning; on the other, learned representations can in turn improve diffusion model performance. A recent survey discusses this bidirectional relationship in depth, providing taxonomies, theoretical underpinnings, and future directions [Li et al. \(2024b\)](#). This perspective supports the representational design and editability objectives of DDAE. DiffuseGAE applies this idea to the latent space of diffusion autoencoders for controllable image manipulation [Leng et al. \(2023\)](#). Self-discovering diffusion [Li et al. \(2024a\)](#) is achieved by identifying vectors in the latent space of diffusion models that are associated with specific concepts. Our approach is complementary: DDAE improves the conditional diffusion decoder and can be combined with the same group-supervised latent editing module.

2.3. Diffusion-Based Image Editing Methods

Diffusion models have demonstrated strong capabilities in image editing. SDEdit leverages the stochastic differential equation (SDE) framework with user input guidance to balance *realism* and fidelity to the source input, without requiring task-specific training or reverse optimization [Meng et al. \(2021\)](#). More broadly, the survey “Diffusion Model-Based Image Editing” provides a systematic taxonomy of current editing methods, covering training strategies, user-controllable conditions, and task types. It also introduces the EditEval benchmark and the LMM (Large Multimodal Model) Score for quantitative evaluation [Xu et al. \(2024, 2025\)](#).

2.4. High-Fidelity Editing and Attention / Flow Control Mechanisms

For precision control and high-fidelity outputs, recent works have proposed architectures that mitigate error accumulation during editing. For instance, High-Fidelity Diffusion-based Image Editing employs a rectifier module to correct cumulative errors in the diffusion process, improving reconstruction and edit quality [Wang et al. \(2023\)](#). Additionally, the “Dual Attention Control” strategy—jointly manipulating cross-attention and self-attention—has been shown effective for high-precision image editing, preserving structural integrity and fine details [Lin et al. \(2025\)](#). This directly inspires the “Dual-Flow” design adopted in DDAE.

3. Proposed Method

3.1. Problem Formulation

Given clean data $x_0 \in \mathbb{R}^N$, a diffusion autoencoder comprises an encoder Enc_ϕ producing semantic code $z_{\text{sem}} = \text{Enc}_\phi(x_0)$, and a conditional decoder D_θ that generates images via iterative denoising. The decoder is trained to predict the noise ϵ_t :

$$\tilde{\theta} = \min_{\theta} \mathbb{E}_{x_0, t} \mathbf{d}[\epsilon_t, \epsilon_\theta(z_{\text{sem}}, x_t, t)] \quad (1)$$

where $\mathbf{d}[\cdot]$ is a similarity metric.

We identify two sources of reconstruction error: (1) unreliable score estimation at certain noise levels, and (2) mismatch between the inversion and generation trajectories.

Target-Dependent Estimation Error. We adopt the variance-preserving (VP) diffusion process with forward noising:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I}) \quad (2)$$

where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$. The posterior mean can be expressed via either x_0 or ϵ_t :

$$q(x_0 | x_t, \epsilon_t) = \frac{1}{\sqrt{\bar{\alpha}_t}} (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_t) \quad (3)$$

The posterior $q(x_{t-1} | x_t, x_0)$ can be estimated using either an x_0 -predictor $x_\theta(\cdot)$ or an ϵ -predictor $\epsilon_\theta(\cdot)$:

$$\mu_x = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_\theta(\cdot); \quad \mu_\epsilon = \frac{1}{\sqrt{\alpha_t}} x_t - \frac{\sqrt{1 - \alpha_t}}{\sqrt{\bar{\alpha}_t}} \epsilon_\theta(\cdot) \quad (4)$$

As shown in Figure 1, these two estimates behave differently across timesteps. At high noise (small t), the signal-to-noise ratio $\lambda_t = \bar{\alpha}_t / (1 - \bar{\alpha}_t) \rightarrow 0$, so x_0 -prediction becomes degenerate—a small ϵ_t error is amplified to arbitrarily large x_0 error. Conversely, at low noise (large t), ϵ -prediction provides weak denoising supervision. The two targets are thus complementary: ϵ -prediction excels at high noise, while x_0 -prediction excels at low noise.

Stochastic Encoder Mismatch. During inference, DDIM inversion maps x_0 to a terminal noise state x_T^* , which is then denoised back to reconstruct the input. The DDIM generation and inversion steps are:

$$\frac{x_{t-1}}{\sqrt{\bar{\alpha}_{t-1}}} = \frac{x_t}{\sqrt{\bar{\alpha}_t}} + \left(\sqrt{\frac{1}{\bar{\alpha}_{t-1}}} - 1 - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \cdot \epsilon_\theta(\cdot) \quad (\text{generation}) \quad (5)$$

$$\frac{x_t^*}{\sqrt{\bar{\alpha}_t}} = \frac{x_{t-1}^*}{\sqrt{\bar{\alpha}_{t-1}}} + \left(\sqrt{\frac{1}{\bar{\alpha}_t} - 1} - \sqrt{\frac{1}{\bar{\alpha}_{t-1}} - 1} \right) \cdot \epsilon_\theta(\cdot) \quad (\text{inversion}) \quad (6)$$

Since the noise predictors in (5) and (6) receive different inputs, their outputs differ, causing trajectory divergence. Assuming $x_t \equiv x_t^*$, the intrinsic gap accumulates as:

$$\Delta = \mathbb{C}_t \cdot \|\epsilon_\theta(z_{\text{sem}}, x_t, t) - \epsilon_\theta(z_{\text{sem}}, x_{t-1}^*, t-1)\| \quad (7)$$

where $\mathbb{C}_t = \sqrt{1 - \bar{\alpha}_t} - \sqrt{1/\alpha_t - \bar{\alpha}_{t-1}}$ depends only on t . This accumulated mismatch degrades high-frequency details during reconstruction.

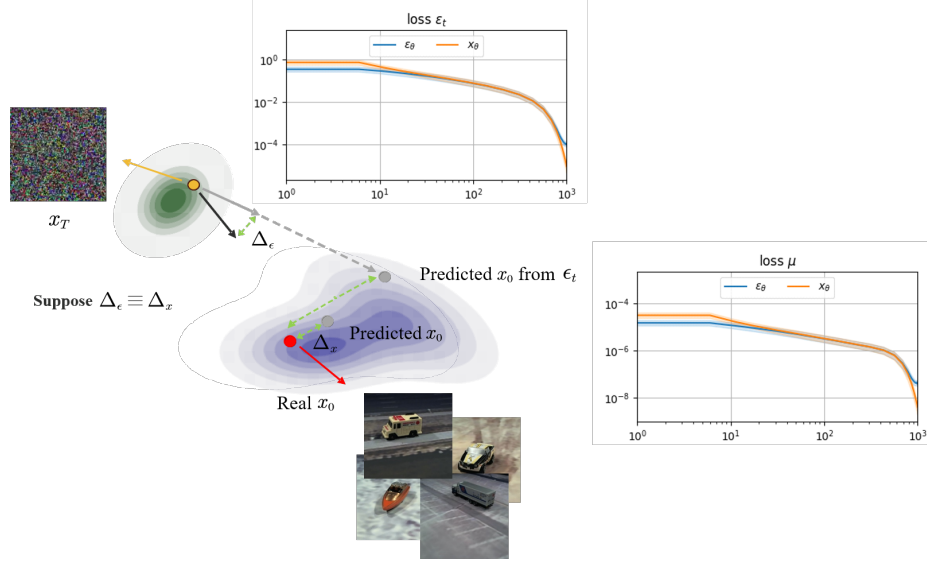


Figure 1: Sensitivity analysis of noise prediction and x_0 prediction across diffusion timesteps. The x-axis denotes timestep t , and the y-axis reports the estimation error on a logarithmic scale. x_0 -prediction yields larger errors than ϵ -prediction at low timesteps (high noise), with this relationship reversed at high timesteps (low noise).

3.2. Model Design

Dual-Flow Adaptive Score Estimation. According to diffusion loss, it can be seen that the simplified training loss function is the optimization of the Kullback-Leibler (KL) divergence between $q(x_{t-1} | x_t, x_0)$ and $p_\theta(x_{t-1} | x_t)$: $L_t := |\mu_t(x_t, x_0, t) - \mu_\theta(x_t, t)|^2 / 2\sigma_t^2$. After training, the generation process $p_\theta(x_{t-1} | x_t)$ of the diffusion probability model can be replaced by the form of the posterior distribution $q(x_{t-1} | x_t, x_0)$, that is:

$$\tilde{\mu}_t(x_t, x_0, t, z_{\text{sem}}) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t; \quad \tilde{\sigma}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t \quad (8)$$

Here, only x_0 is unknown, so the model can predict x_0 to obtain the generated distribution $p_\theta(x_{t-1} | x_t)$. Furthermore, from Equ. (3), it can be inferred that any two of (x_t, ϵ_t, x_0) can predict the remaining one, so x_0 can be replaced by an expression involving x_t and ϵ_t , indicating that the model can also generate by predicting ϵ_t . The posterior generation distribution $q(x_{t-1} | x_t, x_0)$ can be estimated using ϵ_θ and x_θ as shown in the following equation.

$$\begin{aligned} \mu_x &:= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_\theta(z_{\text{sem}}, x_t, t) \\ \mu_\epsilon &:= \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_\theta(z_{\text{sem}}, x_t, t) \end{aligned} \quad (9)$$

Since ϵ_t has zero mean and unit variance, noise prediction implicitly normalizes the model and connects to score-function estimation Song et al. (2021b). Conversely, x_0 -prediction provides an unbiased estimate that is particularly advantageous under heavy corruption, but becomes degenerate at low SNR (early timesteps), approaching a trivial identity mapping. The two parameterizations are therefore *complementary*: ϵ -prediction is preferred at high noise, while x_0 -prediction is preferred at low noise.

We exploit this complementarity with a learned gating function $\gamma_\theta(z_{\text{sem}}, x_t, t) \in [0, 1]$ that adaptively interpolates between μ_x and μ_ϵ at each timestep:

$$\tilde{\mu}_t = (1 - \gamma_t) \mu_x + \gamma_t \mu_\epsilon. \quad (10)$$

The complete pipeline is in Figure 2.

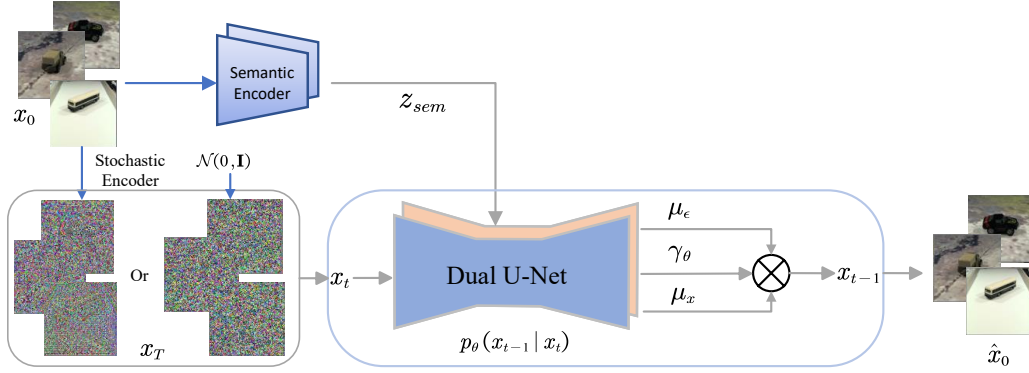


Figure 2: Complete pipeline of the Dual-Flow Adaptive Score Matching algorithm. The model adaptively interpolates between x_0 -prediction and ϵ -prediction paths at each timestep t using a learned gating function γ_θ .

Implicit Constraint in Stochastic Encoder In the diffusion autoencoder, we observe that the key-value features from the self-attention layers contain useful structural information from the input. During DDIM inversion, we cache these features; during generation, we reuse them in selected decoder layers to align local appearance details.

In the diffusion autoencoder, when the stochastic encoder $E_{\phi_2}^t$ (with noise) at different time steps t is used to generate intermediate representations x_t^* with noise, the input of which contains x_{t-1} , it can be considered that the key-value pair features contain additional texture information from the input feature x_{t-1} . When generating the intermediate representation x_{t-1} (denoising) in the generation process, the key-value features of the self-attention layer in the encoder $E_{\phi_2}^t$ can be used in the generation process to guide x_{t-1} towards a direction closer to x_{t-1}^* . This mechanism is illustrated in Figure 3.

It is also worth noting that this constraint is not applied at all timesteps. Because different timesteps and model layers focus on generating different content Cao et al. (2023), applying this constraint in early stages would disrupt the image layout. Therefore, key-value replacement is only applied for $t < t^*$ and $l > l^*$. Empirically, we set t^* to the last 10 steps and l^* to all decoder layers. The complete pseudocode is provided in Appendix 4.

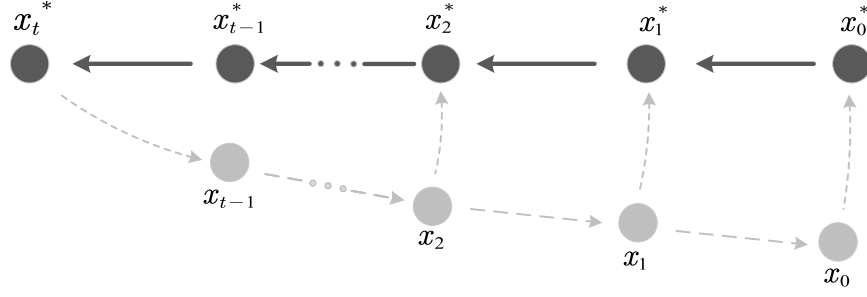


Figure 3: Illustration of the Implicit Constraint mechanism. Key-value attention pairs $\{K_t^*, V_t^*\}$ from the stochastic encoder are injected into the decoder self-attention layers during generation for $t < t^*$ and decoder layers $l > l^*$, guiding the denoising trajectory toward the inverted representation.

3.3. Model Training and Inference

Loss Function In this section, a new loss function L_t is designed for the newly constructed model, which includes three parts for optimizing $\epsilon_\theta, x_\theta$ and γ_θ . Here, $\tilde{\mu}_t, \mu_x(\cdot), \mu_\epsilon(\cdot)$ are solved according to Equ. (11), respectively. The notation $[\cdot]_{sg}$ indicates that the gradient flow during backpropagation is stopped here because $\partial\mu_x/\partial x_\theta, \partial\mu_\epsilon/\partial\epsilon_\theta$ would be affected by the previously mentioned infinite magnitude, hence stopping the gradient flow here makes the training more stable.

$$\begin{aligned}\mathcal{L}_\epsilon &= \mathbb{E}_{\epsilon_t \sim \mathcal{N}(0, \mathbf{I}), t \sim U(0,1)} \mathbf{d}[(\epsilon_\theta(x_t, t, z_{\text{sem}}), \epsilon_t)] \\ \mathcal{L}_x &= \mathbb{E}_{x_0 \sim q(x_0), t \sim U(0,1)} \mathbf{d}[(x_\theta(x_t, t, z_{\text{sem}}), \mathbf{x}_0)] \\ \mathcal{L}_\gamma &= \mathbb{E}_{x_0, t, x_t} \mathbf{d}[\tilde{\mu}_t, \gamma_\theta[\mu_x]_{sg} + (1 - \gamma_\theta)[\mu_\epsilon]_{sg}]\end{aligned}\tag{11}$$

$$\mathcal{L}_{DDAE} = \lambda_\epsilon \mathcal{L}_\epsilon + \lambda_x \mathcal{L}_x + \lambda_\gamma \mathcal{L}_\gamma\tag{12}$$

Here, $\mathbf{d}$ denotes the distance metric, where the prediction for ϵ_t and γ_t still uses the L_2 distance measured in Euclidean space. However, the image space is generally assumed to be a manifold space measured by non-Euclidean distance, thus this section adopts a weighted average of perceptual loss and MSE loss as the final distance metric for x_0 . The perceptual loss is measured using the VGG-16 model, with both λ_p and λ_m set to 1 in this section. See Appendix 5 for more details.

$$\mathbf{d}(\cdot) = \lambda_p \mathbf{d}_p + \lambda_m \mathbf{d}_{mse}\tag{13}$$

Inference The Dual-Flow estimation network after training can also be easily extended to the deterministic generation process of diffusion autoencoders. This section still uses DDIM as the ODE sampler because only the estimation method for $\tilde{\mu}_t$ is changed, which does not affect the ability of implicit sampling. The complete generation procedure is summarized in Algorithm 1. The original DDIM sampling process is as follows:

Input: Image data $x_0 \sim q(x_0)$, Adaptive dual-path diffusion autoencoder $E_{\phi_1}, E_{\phi_2}, D_{\theta}$

Output: High-quality generated image $\hat{x}_0$

Sample noise $x_T \sim \mathcal{N}(0, \mathbf{I})$ or $x_T = \text{Enc}_{\phi_2}(x_0)$

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for  $t = T, \dots, 1$  do
     $\epsilon \sim \mathcal{N}(0, \mathbf{I})$  if  $t > 1$ , else  $\epsilon = 0$ 
    Obtain  $\mu_{\epsilon}, \mu_x$  using Equ. (15)
    Obtain  $x_{t-1}$  using Equ. (16)
end
return  $x_0$ 

```

Algorithm 1: DDAE Generation Procedure

$$\begin{aligned}
 x_{t-1} &= \sqrt{\bar{\alpha}_{t-1}} x_{\text{pred}} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \epsilon_{\theta} + \sigma_t \epsilon_t \\
 \mu_{\text{DDIM}} &:= \sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \epsilon_t
 \end{aligned} \tag{14}$$

Consistent with training, this section’s algorithm also takes the interpolation of μ_x and μ_{ϵ} as the estimate of the mean, yielding the following process for DDAE using a DDIM-like method for generation, with the decoder in this paper being a deterministic process, so σ_t is set to 0:

$$\begin{aligned}
 \mu_{\epsilon} &= \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_{\theta}}{\sqrt{\bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \epsilon_{\theta} \\
 \mu_x &= \sqrt{\bar{\alpha}_{t-1}} x_{\theta} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{x_t - \sqrt{\bar{\alpha}_t} x_{\theta}}{\sqrt{1 - \bar{\alpha}_t}}
 \end{aligned} \tag{15}$$

$$x_{t-1} = (1 - \gamma_t) \mu_x + \gamma_t \mu_{\epsilon} + \sigma_t \epsilon_t \tag{16}$$

4. Experiments

4.1. Experimental Setup

We now turn to assessing the properties of our learned latent space and demonstrating new capabilities, such as attribute manipulation. We put all implementation details in Appendix 6.

Dataset To validate the generalization of the proposed Dual-Flow Adaptive Diffusion Autoencoder (DDAE), we selected datasets from different fields for evaluation, including the vehicle dataset iLab-20M and high-quality facial datasets FFHQ and CelebA-HQ.

4.2. Reconstruction quality

Model comparison on iLab-20M dataset This section begins by assessing the reconstruction quality of the Dual-Flow Adaptive Diffusion Autoencoder (DDAE) on the iLab-20M dataset in comparison to other methods. As shown in Table 1, DDAE-Final shares the entire U-Net backbone and uses separate prediction heads for x_0 and ϵ , while DDAE-Last-3 uses target-specific parameters only in the last three decoder blocks. See Appendix 6.1 for more structure details. On the iLab-20M [Borji et al. \(2016\)](#) vehicle dataset, we found that

Table 1: Reconstruction quality on iLab-20M at 128×128 . $\downarrow$: lower is better; $\uparrow$: higher is better. “Inferred” denotes using the stochastic code obtained by inversion.

Method	FID $\downarrow$	SSIM $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$
GZS-Net (no stochastic code)	100.85	0.345	0.205	0.124
Diff-AE, random x_T , $T = 100$	8.07	0.539	0.073	0.011
DiffuseGAE, random x_T , $T = 100$	8.06	0.549	0.073	0.011
DDAE-Final, random x_T , $T = 100$	7.29	0.617	0.065	0.010
DDAE-Last-3, random x_T , $T = 100$	6.89	0.601	0.061	0.010
DDIM, inferred x_T , $T = 100$	10.37	0.912	0.063	0.002
DiffuseGAE, inferred x_T , $T = 100$	4.03	0.989	0.012	8.56×10^{-5}
DDAE-Last-3, w/o constraint, inferred	3.85	0.989	0.010	8.01×10^{-5}
DDAE-Last-3, w/ constraint, inferred	3.43	0.992	0.009	7.89×10^{-5}

a Symmetric design with largely independent U-Net branches for the two prediction targets performed substantially worse (see ablation in Section 4.3). We therefore focus on the Final and Last architectures, which achieve better or comparable results on all metrics compared to previous methods. The improved results obtained by the DDAE using random noise suggest that during the model generation process, it is able to thoroughly extract information at lower steps and efficiently mine information at higher steps. The Last architecture, featuring more adaptive layers, showed the best performance.

Model comparison on CelebA-HQ dataset To further assess the autoencoding reconstruction capabilities of DDAE, we carried out identical quantitative experiments as those done with Diff-AE. Specifically, we employed “DDAE-Last-3” trained on FFHQ dataset to encode and reconstruct all 30,000 images from the CelebA-HQ dataset, and evaluated the reconstruction quality using their average SSIM Wang et al. (2004), LPIPS Ho et al. (2019), and MSE metrics. We utilized the same baseline comparisons as described in DiffuseGAE, with the outcomes presented in Table 2. Compared to other methods, DDAE exhibits higher reconstruction quality under similar dimensional conditions, highlighting its remarkable data compression efficiency. Figure 4 further shows that DDAE preserves facial geometry and identity under cross-dataset reconstruction.

Figure 4 provides qualitative evidence for the reconstruction results. On iLab-20M, DDAE accurately preserves vehicle identity, viewpoint, and background context while maintaining sharp object boundaries and visually plausible textures. The reconstructions remain close to the inputs even when vehicles appear under different poses and scene layouts, indicating that the inferred terminal noise effectively carries residual high-frequency information beyond the compact semantic code. On CelebA-HQ, the model also reconstructs facial pose, expression, hair structure, and local appearance with high fidelity under the cross-dataset setting where the model is trained on FFHQ and evaluated on CelebA-HQ. Minor smoothing can still be observed in some fine hair and skin details, but the overall identity and facial geometry are well preserved. These visual results support the conclusion that the Dual-Flow

Table 2: Cross-dataset reconstruction on CelebA-HQ using a model trained on FFHQ.

Method	SSIM $\uparrow$	LPIPS $\downarrow$	MSE $\downarrow$
GZS-Net (no stochastic code)	0.345	0.205	0.124
DiffuseGAE, random x_T , $T = 100$	0.677	0.073	0.007
PDAE, random x_T , $T = 100$	0.696	0.094	0.005
DDAE-Last-3, random x_T , $T = 100$	0.602	0.061	0.010
DDIM, inferred x_T , $T = 100$	0.912	0.063	0.002
DiffuseGAE, inferred x_T , $T = 100$	0.989	0.012	8.56×10^{-5}
PDAE, inferred x_T , $T = 100$	0.993	0.008	5.48×10^{-5}
DDAE-Last-3, inferred x_T , $T = 100$	0.995	0.005	4.50×10^{-5}



Figure 4: Qualitative reconstruction results on iLab-20M and CelebA-HQ. For each dataset, the first row shows input images and the second row shows reconstructions produced by DDAE-Last-3 with inferred terminal noise. DDAE preserves object identity, viewpoint, background layout, facial structure, and most high-frequency appearance details, which is consistent with the quantitative improvements reported in Tables 1 and 2.

posterior estimator and the implicit consistency constraint improve reconstruction quality not only numerically but also perceptually.

4.3. Image Editing and Interpolation Trajectories

To validate that the DDAE can achieve high-quality compressed representations, we test the capabilities of DDAE in image editing and image interpolation on the iLab-20M dataset based on the multi-attribute image editing framework DiffuseGAE [Leng et al. \(2023\)](#).

Image Editing We perform image recombination to evaluate the image editing capability of our method. Our method performs image replacement as follows: Given two images, the former provides the part in z_{dis} corresponding to attribute id and pose, while the latter provides the part corresponding to attribute background. Since there is no such target image we wish to synthesize, we adopt $x_T \sim \mathcal{N}(0, \mathbf{I})$ as a noise map to generate images iteratively. Then we pass the $\hat{z}_{sem}$ decoded from recombined $\hat{z}_{dis}$ as condition to generate images with $T = 100$ steps. Figure 5 shows our generative results on image recombination; we can see that our method can effectively improve the quality of generation while ensuring the semantics remain unchanged.

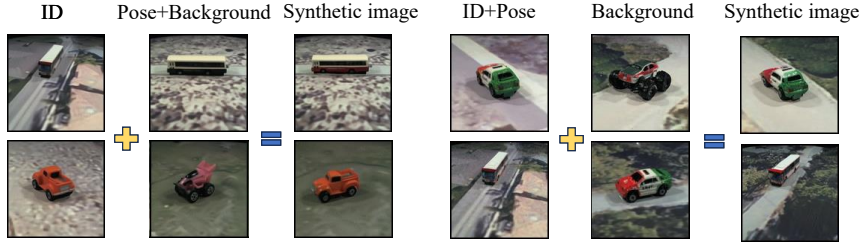


Figure 5: Image editing results on the iLab-20M dataset. Given two source images, DDAE recombines disentangled attributes (id, pose, background) to synthesize new images with high fidelity and semantic consistency.

Interpolation Results To evaluate the smoothness of our method for image interpolation, we present interpolation examples on single and multi-attribute cases. Given two images $x_0^{(1)}$ and $x_0^{(2)}$ from dataset $\mathcal{D}$, We encode them into $(z_{dis}^{(1)}, x_T^{(1)})$ and $(z_{dis}^{(2)}, x_T^{(2)})$ using our trained model. We evaluate our method qualitatively and quantitatively by running our generative process of conditional DDIM from $Slerp(x_T^{(1)}, x_T^{(2)}; \alpha)$ under the condition $\hat{z}_{sem}$ with 100 steps, expecting smooth trajectories in the direction along α . The $\hat{z}_{sem}$ is decoded from $Lerp(z_{dis}^{(1)}, z_{dis}^{(2)}; \alpha)$ by GAE from [Leng et al. \(2023\)](#). We only work on the part of z_{dis} which corresponds to the attribute we desire to interpolate. Figure 6 shows the trajectories for various values of $\alpha \in [0, 1]$ among three attributes. Our methods produce smooth results with well-preserved details from both images and also hold robust semantic consistency, i.e., only changing what we want, which verifies the disentanglement performance of our method.

To quantify the smoothness of the interpolation, we adopt the Perceptual Path Length (PPL) introduced in StyleGAN [Karras et al. \(2021\)](#). In particular, we compute the following expectation over randomly sampled 1k pairs of latent codes (z_1, z_2) and $\alpha \sim U(0, 1)$. As shown in Fig. 7, our method outperforms others under all timesteps in terms of interpolation smoothness.

4.4. Ablation Study

We ablate the key components of the proposed method, including the Dual-Flow score estimation, the implicit consistency constraint, and the network architecture.



Figure 6: Interpolation trajectories on the iLab-20M dataset for $\alpha \in [0, 1]$ across three attributes. DDAE produces smooth transitions with well-preserved details and robust semantic consistency.

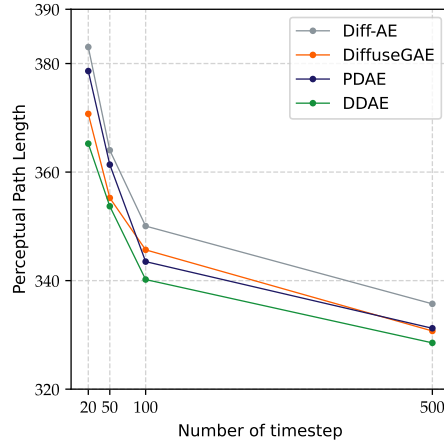


Figure 7: Perceptual path length ($\downarrow$) for various timesteps on CelebA-HQ dataset.

Dual-Flow Score Estimation. Table 3 presents ablation results on the iLab-20M dataset across different sampling steps $T \in \{10, 20, 50, 100\}$. We evaluate three key variants:

- **Only μ_ϵ :** Using noise-prediction exclusively with $\gamma_t = 1$.
- **Only μ_x :** Using x_0 -prediction exclusively with $\gamma_t = 0$.
- **Fixed γ_t :** Using a fixed gating ratio, which was taken to be the mean value for the interpolation parameter γ_θ over the generation steps.

The results validate our design choices:

- **Single-target limitations:** Using only μ_ϵ yields the worst performance across all T (FID 21.53 at $T = 10$), confirming that noise prediction alone is unreliable at low noise

Table 3: FID scores ($\downarrow$) for unconditional generation on FFHQ.

Model	FID $\downarrow$			
	$T = 10$	$T = 20$	$T = 50$	$T = 100$
DDIM	29.56	21.45	15.08	12.03
Diff-AE	20.80	16.70	12.57	10.59
PDAE	20.16	17.18	12.81	10.31
DDAE (full)	18.06	11.54	8.54	8.07
- fixed γ_t	19.24	11.85	9.20	8.23
- only μ_ϵ	21.53	18.64	13.53	10.68
- only μ_x	19.53	12.18	9.57	8.45

levels (high timesteps). Conversely, using only μ_x (FID 19.53 at $T = 10$) outperforms μ_ϵ -only at small T but degrades at large T (FID 8.45 at $T = 100$), validating that x_0 -prediction fails under heavy noise.

- **Learned vs. fixed gating:** The fixed gate provides a reasonable baseline (FID 19.24 at $T = 10$) but still lags behind the learned gate (FID 18.06 at $T = 10$). This confirms that adaptively weighting μ_ϵ and μ_x based on both timestep and semantic latent provides additional benefit over a simple linear schedule.
- **Full Dual-Flow:** The complete DDAE with learned gate achieves the best results across all sampling steps, demonstrating that adaptive fusion effectively leverages the complementary strengths of the two prediction targets.

Implicit Constraint. As shown in Figure 8, models with the implicit constraint consistently outperform those without across different architectures and sampling steps, validating that aligning the inversion and generation trajectories improves generation quality.

Model Architecture. We ablate three network architecture designs to study parameter sharing between the x_0 -prediction and ϵ -prediction flows. The Symmetric design assigns largely independent U-Net branches to the two targets, the Final-head design shares the whole U-Net trunk and only separates the final prediction heads, and the Last- k design shares the encoder, bottleneck, and early decoder layers while using target-specific parameters in the last k decoder blocks.

The Symmetric design performs substantially worse, indicating that the two prediction targets should not be modeled by two almost independent denoising networks. Although x_0 and ϵ have different numerical properties, they are derived from the same noisy image and semantic condition, and therefore require shared low-level and mid-level denoising representations. The Final-head design is more stable because most representations are shared, but its target-specific capacity is limited to the output projection. In contrast, Last-3 provides a better trade-off: it preserves shared denoising features in the early and middle layers while allowing the last decoder blocks to specialize. We use Last-3 as the default DDAE architecture.

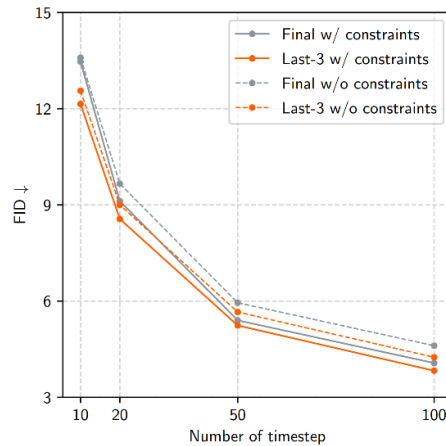


Figure 8: FID comparison on iLab-20M with and without the implicit constraint under different architectures and timestep settings. Implicit constraints consistently improve generation quality.

5. Conclusion

We present the Dual-Flow Adaptive Diffusion Autoencoder (DDAE), an image generation algorithm leveraging decoupled representations and adaptive Dual-Flow score matching to surpass traditional models in generating high-quality, realistic images. At its heart, DDAE enhances score matching accuracy through a novel autoencoder architecture and adaptive matching, while implicit constraints on random encoders boost image generation consistency and stability. Experiments on controlled vehicle data and face data show improved reconstruction fidelity and perceptual quality, while latent-space recombination and interpolation demonstrate compatibility with disentangled editing. The addition of implicit constraints further elevates generation quality. In summary, DDAE offers an effective approach to image generation, with significant potential for improving image quality, model stability, and broadening application horizons. We believe it will be a promising tool for more applications in both image and video generation, which we point to as future work.

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